

ROBERT (ROSS) WARDRUP

Spatial & Computational Epidemiology | MPH Epidemiology Candidate | Bayesian Spatial Models
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SUMMARY

MPH Epidemiology candidate and public health data professional with a background in medical geography and over a decade of experience developing and analyzing large health and geospatial datasets. Current graduate research applies Bayesian spatial models to vector-borne disease surveillance, with an emphasis on simulation-based model validation and calibration. Professional experience spans public health surveillance, healthcare analytics, and federal geospatial hazard modeling, using SQL, R, Python, and GIS technologies. Sole- and co-author on peer-reviewed and preprint epidemiologic research. Seeking research and doctoral opportunities in spatial and computational epidemiology.

RESEARCH INTERESTS

Spatial epidemiology; computational epidemiology; Bayesian spatial and spatiotemporal modeling; infectious and vector-borne disease epidemiology; environmental health; health disparities; maternal and child health; health services research.

THESIS RESEARCH

MPH THESIS: BAYESIAN SPATIOTEMPORAL MODEL OF WEST NILE VIRUS (CALIFORNIA) Aug 2022 – Present
University of Memphis Remote

- Designing a Bayesian hierarchical model (pooled-prevalence, BYM2-analog) of West Nile virus vector positivity across California counties, 2006–2024, with distributed-lag climate covariates and queen-contiguity spatial structure.
- Because the primary surveillance dataset is restricted and gated pending IRB approval, built a simulation-based validation framework first: real public covariate structure, simulated outcomes with a known effect, then a check of whether each candidate model's 95% intervals covered that effect at the stated rate.
- Evaluated interval coverage across dozens of model specifications and simulated data-generating scenarios varying temporal autocorrelation, overdispersion, and disease positivity; found that omitting seasonal and month-to-month temporal structure caused substantial under-coverage of uncertainty intervals, while point estimates alone did not reveal the problem.

TECHNICAL SKILLS

Epidemiology & Research: descriptive epidemiology; retrospective cohort studies; surveillance analysis; health disparities research; longitudinal data analysis; regression modeling; Bayesian hierarchical modeling; simulation studies and model calibration; spatial epidemiology; data validation; reproducible research

Programming & Data Management: SQL (PostgreSQL/PostGIS, T-SQL); R (tidyverse, ggplot2, R-INLA); SAS; Python (pandas, NumPy, ArcPy); Git

Clinical & Public Health Data: healthcare datasets; public health surveillance data; database development; ETL workflows; data quality assurance; documentation

Visualization & GIS: Tableau; ArcGIS Pro; ArcGIS Online; QGIS; FME

EDUCATION

UNIVERSITY OF MEMPHIS Remote
Master of Public Health (MPH), Epidemiology (in progress) Aug 2022 – Present

- Graduate coursework: Epidemiology I & II, Biostatistics I & II, Environmental Epidemiology, Spatial Epidemiology, Epidemiologic Methods, Survey Research Methods, Causal Inference, Environmental Health.

UNIVERSITY OF NORTH TEXAS Denton, TX
Bachelor of Science in Geography May 2013

PUBLIC HEALTH EXPERIENCE

MPH EPIDEMIOLOGY PRACTICUM INTERN May 2026 – Present
Southeast Utah Health Department Remote / Price, UT

- Analyzed 3,454 respiratory illness records (2015–2026) with ArcGIS Pro/Python, Getis-Ord Gi*, and Emerging Hot Spot Analysis to describe spatial and temporal disease burden across the jurisdiction.
- Produced reusable surveillance maps, method notes, and briefing materials used in SEUHD planning discussions, including mobile vaccine clinic planning.

- Built EHR-based (eClinicalWorks/eBO) immunization compliance logic for school-entry requirements, using CVX-to-requirement mapping, age-gated criteria, distinct-dose counting, and missing-dose flags.
- Identified a hidden EHR report-duplication issue that was inflating vaccine dose counts and authored QA documentation to prevent its use in outreach before correction.

EPIDEMIOLOGY AND DATA SCIENCE PROFESSIONAL (TEMP)

January 2021 – May 2021

Tri-County Health Department

Remote

- Developed ETL workflows using FME and FME Server to integrate public health surveillance datasets into centralized geospatial databases.
- Designed and maintained PostGIS databases supporting COVID-19 case surveillance and vaccination monitoring efforts.
- Performed quality assurance and validation of public health datasets to improve reliability and consistency of analytic outputs.

PROFESSIONAL EXPERIENCE

GIS DEVELOPER (CONTRACT)

December 2024 – Present

FEMA via Marathon TS and IBM

Remote

- Led a 2,000-event validation study comparing two hazard-data source formats for a national disaster-risk model, using descriptive statistics and confidence intervals to quantify agreement and identifying a geospatial data-processing defect that was corrected prior to deployment.
- Designed a custom Python geospatial data abstraction standardizing coordinate reference handling, spatial joins, and feature-class I/O across hazard modules of a federal risk-modeling application.
- Maintain SQL Server databases supporting federal hazard-modeling workflows, including scripted quality-assurance checks to support reproducible model execution.

GIS ANALYST

May 2023 – November 2024

DaVita Kidney Care

Remote

- Developed Python and ArcPy workflows supporting healthcare operations and resource allocation across a national dialysis provider network.
- Conducted spatial and statistical analyses of healthcare access, service distribution, and patient-care delivery patterns using R and GIS tools.

Prior GIS Roles: Staff GIS Developer, Axim Geospatial (Dec 2021 – May 2023) — geospatial planning/mapping application development (Python, VB.NET, JavaScript/TypeScript) and QGIS data validation. GIS Engineer, Congruex / CHC Consulting (May 2021 – Dec 2021) — fiber-network infrastructure design and utility spatial data analysis.

SELECTED PROJECTS

Spatial Analysis of Vaccination Exemptions in Colorado

Nov 2023

- Investigated spatial clustering of non-medical vaccination exemptions across Colorado using hotspot analysis and spatial interpolation.
- Applied kriging and rate-smoothing techniques in R to characterize geographic variation in exemption rates and evaluate associations with socioeconomic indicators.

PUBLICATIONS

THE IMPACT OF COVID-19 ON GLOBALIZATION

Dec 2020

One Health, 11:100180

doi: 10.1016/j.onehlt.2020.100180

- Contributed quantitative analysis examining relationships between pandemic spread, mobility patterns, and global economic disruption.

SPATIAL PATTERNS OF COVID-19 MORTALITY: EXAMINING SOCIOECONOMIC DETERMINANTS IN U.S. COUNTIES USING CLUSTER ANALYSIS

Preprint, medRxiv

2024

- Sole-author study using cluster analysis to identify spatial patterns in U.S. county-level COVID-19 mortality and examine associated socioeconomic determinants.

PROFESSIONAL MEMBERSHIPS

SOCIETY FOR EPIDEMIOLOGIC RESEARCH (SER)

May 2025 – Present

Student member

COUNCIL OF STATE AND TERRITORIAL EPIDEMIOLOGISTS (CSTE)

May 2025 – Present

Student member